

- 3 The photograph shows a fruit fly, *Drosophila suzukii*, on the skin of a fruit.



(Source: © dpa picture alliance/Alamy Stock Photo)

This species of fruit fly is endemic to southeast Asia but recently it has become an invasive species in Europe.

The female lays fertilised eggs under the skin of fruits.

The eggs hatch into larvae that feed on the fruit.

The larvae develop into pupae that become adult flies.

A student investigated the effect of using an organic pesticide, pyrethrum, on the hatching of eggs and the development of pupae.

Sixteen flasks containing a culture medium were prepared. Eight flasks had 0.5 g of pyrethrum added.

Each flask had one female fly that laid eggs for 24 hours. The fly was then removed and the eggs were allowed to hatch and develop into pupae.

The photograph shows fly pupae in one of the flasks.



(Source: © Sundry Photography/Alamy Stock Photo)

After five days, the number of pupae in each flask was counted.

*The results were:*

*Number of pupae from untreated culture medium (A)*

72 68 81 56 43 52 60 64

*Number of pupae from treated culture medium (B)*

45 56 39 40 29 38 35 46

(a) State a suitable null hypothesis for this investigation.

(1)

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(b) Draw a suitable table to display these **data** and the calculated **means** for the number of pupae from the flasks containing the untreated and treated culture medium.

(2)

(c) Draw a suitable graph to show the mean number of pupae from the untreated culture medium and treated culture medium.

Include an indication of the variability of the data.

(3)



(d) The student analysed the data with a  $t$  test using the formula:

$$t = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

where:

$\bar{x}$  is the mean value for each treatment

$n$  is the number of samples for each treatment

$(S_A)^2 = 143.1$  and  $(S_B)^2 = 65.7$

(i) Calculate the value of  $t$ .

(2)

Answer .....

(ii) The table shows the critical values of  $t$  for different degrees of freedom.

The number of degrees of freedom =  $(n_A - 1) + (n_B - 1)$

| Degrees of freedom | p = 0.05 | p = 0.01 |
|--------------------|----------|----------|
| 10                 | 2.23     | 3.17     |
| 11                 | 2.20     | 3.11     |
| 12                 | 2.18     | 3.05     |
| 13                 | 2.16     | 3.01     |
| 14                 | 2.14     | 2.98     |
| 15                 | 2.13     | 2.95     |
| 16                 | 2.12     | 2.92     |
| 17                 | 2.11     | 2.90     |
| 18                 | 2.10     | 2.88     |
| 19                 | 2.09     | 2.86     |
| 20                 | 2.09     | 2.84     |
| 21                 | 2.08     | 2.83     |
| 22                 | 2.07     | 2.82     |

Deduce the conclusions that can be drawn from this investigation.

Use the information in this table to support your answer.

(2)

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(e) Describe **two** ways this investigation could be extended to collect more data to either support or reject the hypothesis.

(2)

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(f) Pyrethrum is applied to fruit crops growing in fields.

Suggest **two** reasons why applying pyrethrum to fruit crops in fields might not reduce damage to fruits.

(2)

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(Total for Question 3 = 14 marks)